In-situ Investigations of CH_4 Decomposition on Nickel Silicate: Switching from Tip to Base Growth of Carbon Nanotubes

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Catalytic CH_4 decomposition is a promising route for the CO_2-free production of H_2 and carbon nanotubes (CNTs) of high added value. Several transition metals have been extensively studied, with Ni-based catalysts being among the most active [1].

Nevertheless, no commercial process has been established to date. The main challenge associated with the typical tip-growth mechanism is the difficulty in regenerating the catalyst. Thus, the so-called base growth mechanism is fundamental for avoiding detachment of the active sites prior to any regeneration.

Metal phyllosilicates are optimal catalysts precursors. Under certain thermal conditions, the ex-solution process leads to finely dispersed nanoparticles with strong metal-support interaction. Surprisingly, these materials have not been fully explored for the CH_4 decomposition reaction [2,3].

In this work, we employed nickel phyllosilicate (Ni-PS) as an effective precatalyst for the base growth of carbon nanotubes. These materials are composed of Ni^{2+} sites allocated on an octahedral coordination between layers of SiO_2 of tetrahedral configuration (Figure 1). After thermal treatment, strongly bonded Ni particles are formed. For the first time, the combination of DFT calculations and *in-situ* XRD studies provide insights into the reducibility of Ni-O bonds in the nickel phyllosilicate, revealing that the high temperatures required for its reduction are related to the reduction of oxygen bonded to silicon atoms (apical oxygen). Moreover, thermodynamical calculations and environmental ETEM suggest that the nanoparticles formation starts by reduction of hydroxyl groups attached to Ni atoms (Ni-OH), followed by their growth described by an autocatalytic process.

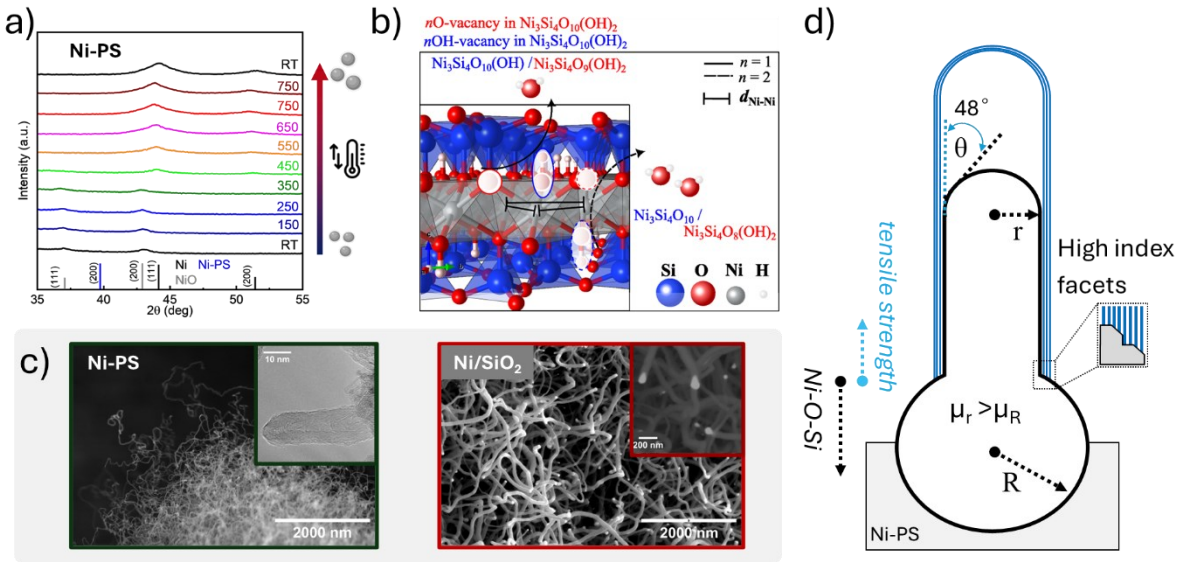

Figure a) In-situ XRD under 10% H_2 for Ni phyllosilicate (Ni-PS) describing NiNPs formation, **b)** Modelled Ni-PS and DFT calculations on oxygen vacancies formation and H_2O formation, **c)** SEM micrographs after CH_4 decomposition and CNTs formation at 550°C for Ni-PS and impregnated Ni/SiO_2, **d)** Schematic representation of the base growth mechanism driven by the Ni-O-Si metal-support interaction described by the experimental evidence in this work.

After activation, *operando* XAS and *in-situ* XRD proved that the small and dispersed Ni^0 nanoparticles act as active sites for CH_4 decomposition. Surprisingly, the particles remain attached to the support, switching from the typical tip-growth for state-of-the-art Ni/SiO_2 impregnated catalyst, to the base growth mechanism for Ni-PS. Contrary to previous findings, it was ruled out the formation of Ni_3C species within the bulk. Furthermore, and disagreement with the current literature state, it is proved that Ni particles below 10 nm can activate CH_4, avoiding encapsulation, and promoting the growth of micrometer-scale, narrow carbon nanotubes.

The formed Ni nanoparticles remained stable against sintering even at temperatures up to 750°C. These findings are highly relevant not only for base-growth carbon nanotube production from CH_4, but also opening a pathway for the rational design of tailored transition metal-based catalysts with high stability under harsh thermal conditions

References

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